



VIAVI

VIAVI Solutions



AEA International:

# OSPREY Training

March 2026

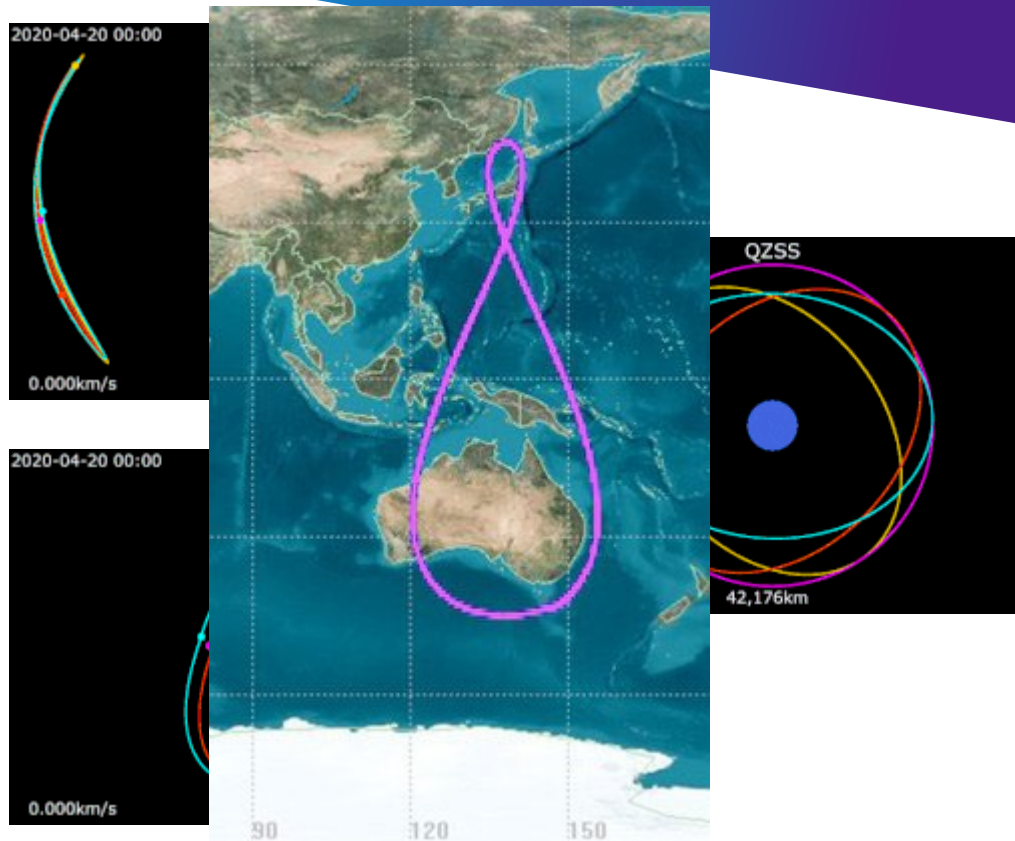
**Mark Anderson, Applications Engineering**

# GPS Overview

- Who does what now?

## Constellation Breakdown:

- GPS (USA) : 31
- GLONASS (Russia): 24
- Galileo (European Union): 25
- BeiDou/BDS (China): 44
- NavIC (India): 7
- QZSS (Japan): 4



# GPS Overview

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## Orbits:

- [LEO \(~1200 miles\)](#)
- [MEO \(~12550 miles\)](#)
- [GEO \(22,236 miles\)](#)

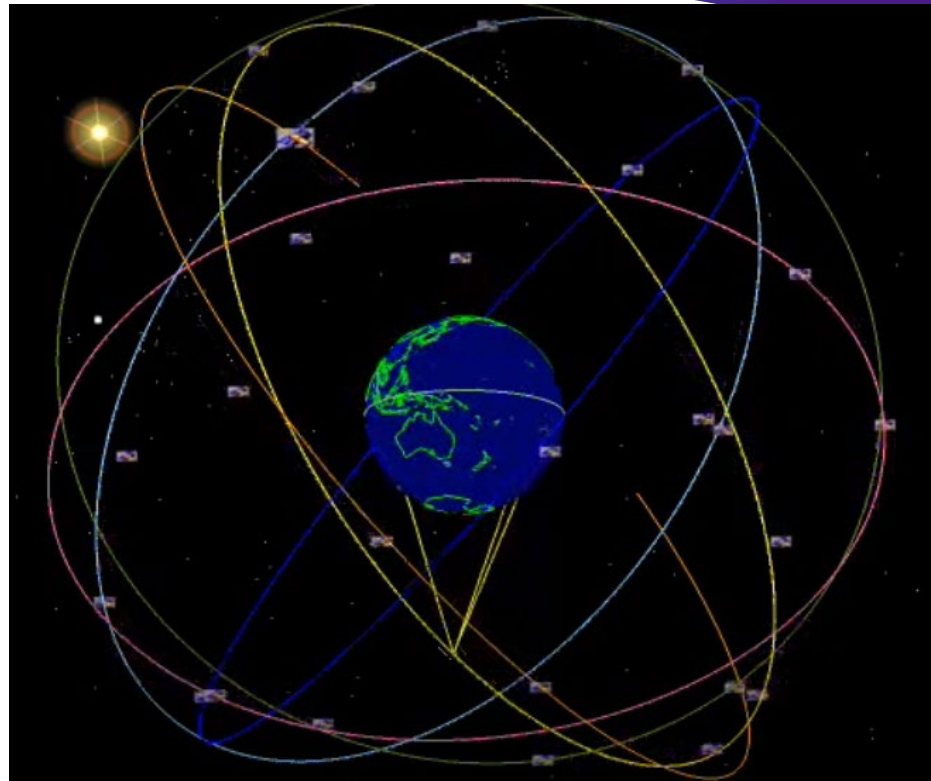
GPS Satellites operate in Medium Earth Orbit while most other space equipment (Starlink: 7,875) operate in Low Earth Orbit

# Almanac Data

- What is it and why do I need it?

A reduced set of data that provides azimuth and elevation data for all of the Space Vehicles in the GPS constellation.

An almanac file provides spatial location information at a given point in time that simulators use as a baseline. Locations are then extrapolated, however, the longer the timeframe, the lower the accuracy, the less likely you are to obtain receiver GPS lock.



# Almanac Data

- What is it and why do I need it?

OSPREY utilizes a RINEX file from NASA.

## RINEX data provided daily/hourly by NASA:

```
1 25 5 5 0 0 0.0 0.248158350587D-03 0.140971678775D-10
0.000000000000D+00 0.239000000000D+03 0.390000000000D+02
0.448697261471D-08 -0.242292608940D+00 0.211782753468D-05
0.515963882208D-03 0.725127756596D-05 0.515372778511D+04
0.864000000000D+05 -0.186264514923D-08 0.224562023971D+01-
0.279396772385D-07 0.959505714281D+00 0.240218750000D+03
0.144979004260D+00-0.809712299196D-08 0.101075638777D-09
0.100000000000D+01 0.236500000000D+04 0.000000000000D+00
0.200000000000D+01 0.000000000000D+00-0.931322600000D-08
0.751000000000D+03 0.842100000000D+05 0.400000000000D+01
0.000000000000D+00 0.000000000000D+00
```

RINEX: Receiver Independent Exchange Format



# OSPREY GPS Simulator

- **O**perational **S**imulated **P**ositioning for the **R**real-time **E**valuation of s**Y**stems

VIAMI's GPS Simulator provides static waypoints and dynamic routes to be simulated for avionics testing purposes.

Daily Almanac data can be seamlessly uploaded to the OSPREY, simplifying the process of ensuring a position lock for GPS Receivers.



# OSPREY Placement

- Single and Dual Coupler Options
- The single coupler option can be used to test pilot and copilot GPS systems individually with the alternate GPS system disabled.
- The dual coupler option can be used to test both pilot and copilot GPS systems simultaneously.
- Direct connection to the LRU is also available as an option.



# Mobile Application

- So easy a caveman can do it...



Fig 1: Remote application home screen



Fig 2: Remote application with static waypoints

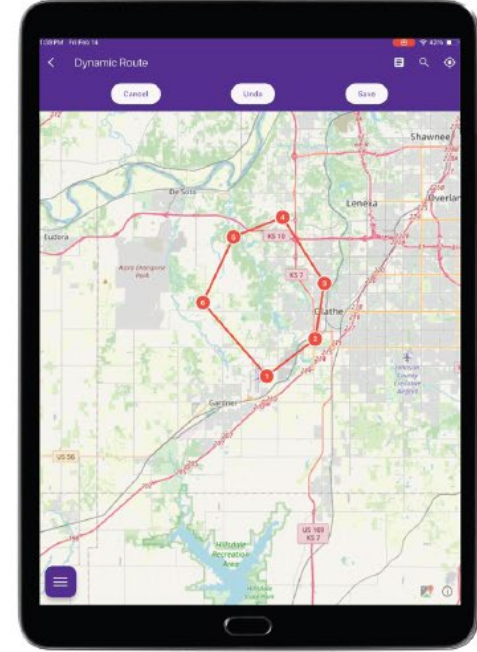
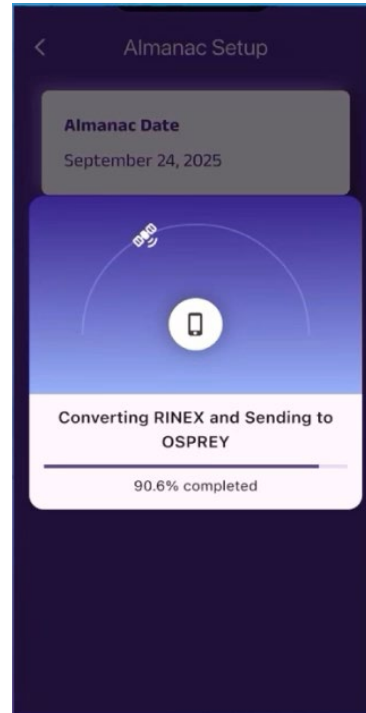


Fig 3: Remote application with dynamic waypoints



# Mobile Application

- So easy a caveman can do it...



# The ABCs (and more) of GPS Receiver 3D Lock

- ✓ A – Almanac Update!
- ✓ B – Be close to last known position
- ✓ C – Close the doors! Isolate from real satellite signals
- ✓ D – Don't need WAAS!
- ✓ E – Ensure power setting is adequate
- ✓ F – First, simulate, then apply power to GPS Receiver

# GPSG-1000 and OSPREY

## Product Comparison

| Use Cases                                            | OSPREY | GPSG-1000 |
|------------------------------------------------------|--------|-----------|
| AC 20-185B and 91.227 ADS-B testing                  | *      | *         |
| RAIM                                                 | *      | *         |
| Dynamic Routes                                       | *      | *         |
| Static Positions                                     | *      | *         |
| Standard Dual Antenna Testing                        | *      | *         |
| GPS Changeover Testing                               | *      | *         |
| TAWS                                                 | *      | *         |
| TCAS                                                 | *      | *         |
| GBAS                                                 | *      | *         |
| LPV                                                  |        | *         |
| FMS                                                  | +      | *         |
| Flight Director                                      | +      | *         |
| Autopilot                                            | +      | *         |
| Full GPS Certification and Engineering Level Testing |        | *         |
| Avionics Integrated with GPS                         | *      | *         |
| L1, L1C, L2C, L5, E1, E5, E5a, and E5b               | L1     | *         |
| Channels simulated                                   | 10     | 12        |

Note: Use cases marked with a + will have some vertical limitations until the OSPREY is able to simulate altitude changes.



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The background of the slide is a high-angle aerial photograph of a vast, arid desert landscape. The terrain is a mix of tan and brown hues, with some darker patches of vegetation or water visible. In the upper portion of the image, the horizon line of the Earth is visible against a deep blue sky. Overlaid on the left side of the image is a glowing, digital wireframe model of a fighter jet. The model is composed of numerous bright blue and white points connected by thin lines, creating a transparent, mesh-like structure. The jet is shown in a dynamic, banking position, with its wings and tail fins clearly defined by the wireframe. Several bright, glowing beams of light emanate from various points on the jet's structure, extending outwards across the sky and over the desert. The overall composition suggests a theme of advanced technology, aerospace, and digital innovation.

**Thank you!**

# GPS Overview

- Cool website

Launch LeoLabs Animation



# GPS Overview

## Terminology

### GPS and AVIATION RELATED ACRONYMS

SV

SBAS

WAAS

RNAV

LNAV

VNAV

LPV



# GPS Overview

## Terminology

### GPS and AVIATION RELATED ACRONYMS

SV



SBAS

WAAS

RNAV

LNAV

VNAV

LPV

Space Vehicles: Individual satellites that make up the GPS constellations



# GPS Overview

## Terminology

### GPS and AVIATION RELATED ACRONYMS

SV

SBAS



WAAS

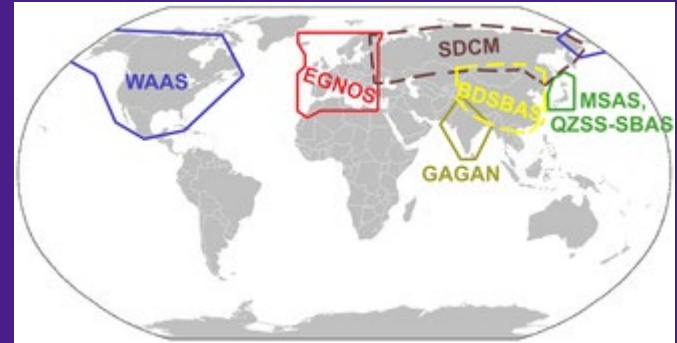
RNAV

LNAV

VNAV

LPV

Satellite Based Augmentation System:  
A general term for systems like WAAS



# GPS Overview

## Terminology

### GPS and AVIATION RELATED ACRONYMS

SV

SBAS

WAAS



RNAV

LNAV

VNAV

LPV

Wide Area Augmentation System:  
Improves accuracy, integrity, availability of GPS  
(Geostationary SVs)

Oh, by the way:  
GPS Accuracy without WAAS – 30 meters  
GPS Accuracy with WAAS – 3 meters

# GPS Overview

## Terminology

### GPS and AVIATION RELATED ACRONYMS

SV

SBAS

WAAS

RNAV

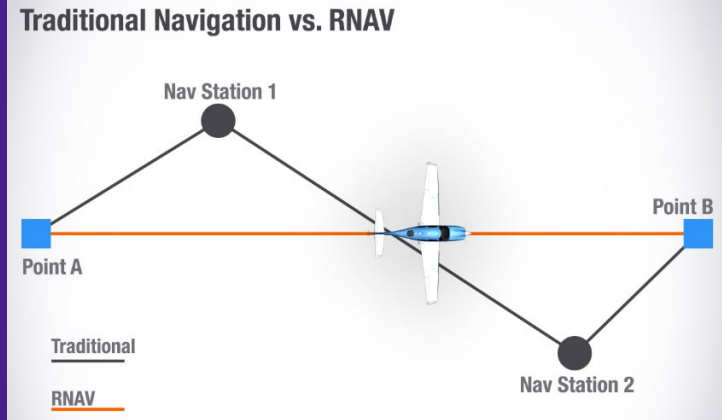
LNAV

VNAV

LPV



Area Navigation: Enables aircraft to navigate directly from one point to another, rather than being restricted to routes defined by ground-based navigation beacons.



# GPS Overview

## Terminology

### GPS and AVIATION RELATED ACRONYMS

SV

SBAS

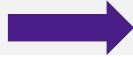
WAAS

RNAV

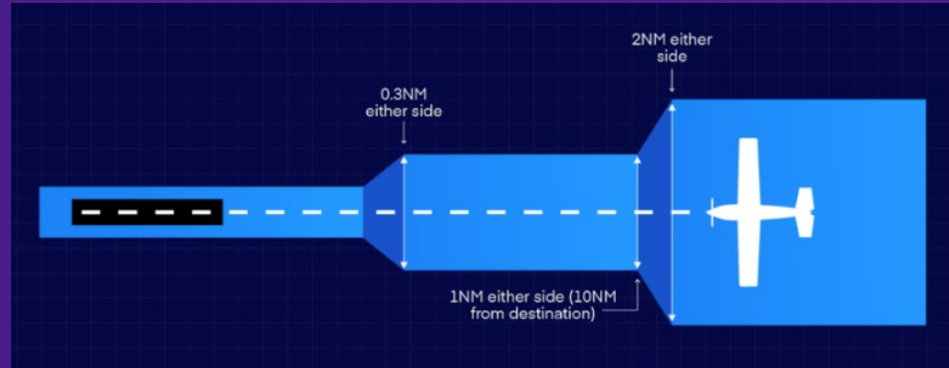
LNAV

VNAV

LPV



Lateral Navigation: Guides aircraft along a horizontal path (think Localizer)





# GPS Overview

## Terminology

### GPS and AVIATION RELATED ACRONYMS

SV

SBAS

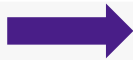
WAAS

RNAV

LNAV

VNAV

LPV



Vertical Navigation: Guides aircraft along a vertical path (think Glideslope). Combines with LNAV for approaches.

Lateral Navigation and Vertical  
Navigation (LNAV/VNAV)



# GPS Overview

## Terminology

### GPS and AVIATION RELATED ACRONYMS

SV

SBAS

WAAS

RNAV

LNAV

VNAV

LPV



Localizer Performance with Vertical Guidance:  
GPS Instrument approach leveraging the accuracy  
of WAAS

Localizer Performance with  
Vertical Guidance (LPV)



# GPS Overview

Spoofing – Is it all bad?

# GPS Overview

## Spoofing – Is it all bad?

- Yes.



# GPS Overview

## Spoofing – Is it all bad?

- Yes. GPS Spoofing is a term that implies a malicious attempt to trick a GPS receiver into believing a false location. Also see GPS Jamming.

- GPS Simulation? ... Good! ★

A testing method that uses a controlled environment to mimic real GPS signals for evaluating GPS receiver performance, security, accuracy, and sensitivity.

# Simulation Types

## Static vs. Dynamic vs. Trajectory

Static Waypoints – probably >80% of use cases

- Simulates one non-moving location's Latitude / Longitude / Altitude

Dynamic Routes

- A series of a number of waypoints that make up a flight / travel path

Trajectory Path (can include Record / Playback capability)

- Similar to Dynamic but uses high throughput NMEA data as the input. NMEA data is a standard protocol for information transfer. The commands look similar to this:

```
$GPGGA,123519,4807.038,N,01131.000,E,1,08,0.9,545.4,M,46.9,M,,*47
```



# Benefits of GPS Simulation

## When and Why?

Simulate numerous locations in and around your hangars

Verification of system installations

Efficient ADS-B testing

Train technicians on new avionics installs

Train pilots on upgraded avionics without flying the aircraft

Testing XM Weather systems

No fines incurred (See SAFO 24002)

# Benefits of GPS Simulation

## GPS Related Aircraft Systems

GPS technology now touches a lot of avionics-related equipment:

- Transponder w/ ADS-B
- Autopilot
- Traffic Collision Avoidance Systems (TCAS)
- Datalink
- Flight Director
- Synthetic Vision Systems (SVS)
- Terrain Awareness and Warning System (TAWS)
- Weather Systems

Consider the possibilities!

# Use Cases

Anywhere GPS Solutions are needed

Use a static location to simulate position inside a hangar.

Eliminates dead spots and the need to shuffle planes.

Stay indoors in bad weather!

Repeaters are typically amplified and can cause issues when doors are open and you are near an active runway.



# Use Cases

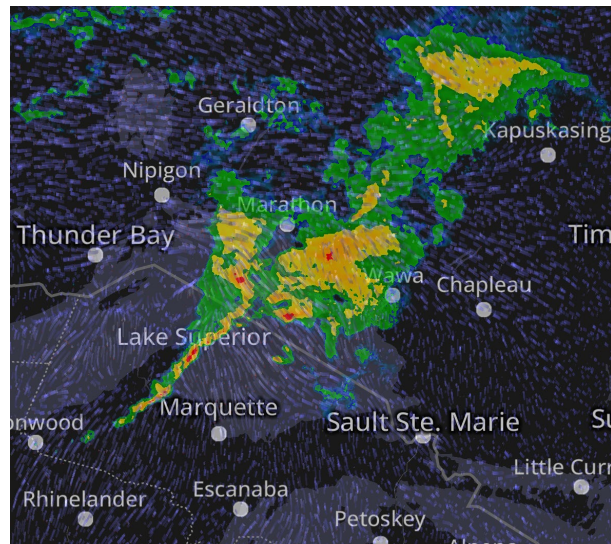
Anywhere GPS Solutions are needed

Use a static location to simulate position where there is bad weather.

Find a location where there is currently some inclement weather and determine the location coordinates.

Simulate that position to test XM Weather System

NOTE: May require a cold-start of the GPS Receiver.



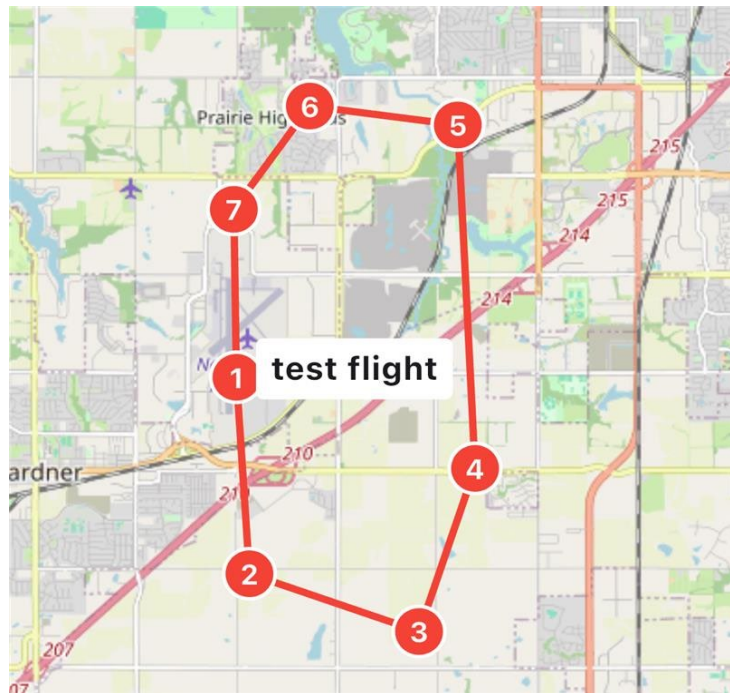
# Use Cases

Anywhere GPS Solutions are needed

Use a dynamic routes to “fly” a pattern.

Fly test patterns for pilot training purposes on new cockpit installs.

Use for creating TA and RA when testing Target Generation



# VIAVI GPS Solutions

## GPSTG-1000 and OSPREY



# VIAVI GPS Solutions

## GPSSG-1000 and OSPREY

| Feature                                          | OSPREY                                                                                                                                       | GPSSG-1000                                                                  |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Number of channels                               | 10                                                                                                                                           | 12                                                                          |
| GPS Signals                                      | GPS L1                                                                                                                                       | GPS L1, L1C, L2C, L5<br>Galileo E1, E5, E5a, E5b                            |
| WAAS Capable                                     | No                                                                                                                                           | Yes                                                                         |
| Static and Dynamic support                       | Yes, with limited dynamics (fixed altitude/speed)                                                                                            | Yes, plus recording and playback of trajectories                            |
| Programmable Space Vehicle parametric and health | No                                                                                                                                           | Yes                                                                         |
| Remote Control                                   | via Phone app                                                                                                                                | Remote Command Interface via ethernet port                                  |
| Power Output Control for Sensitivity testing     | Yes                                                                                                                                          | Yes                                                                         |
| Almanac download                                 | Super simple via phone application<br>Less than 2 minutes                                                                                    | Supports live satellite feed download (15 minutes) or via USB file transfer |
| Typical Use cases                                | Replacing GPS Repeaters<br>Anything where ADS-B is needed<br>Fly routes for pilot training after cockpit upgrade<br>XM Weather Radar testing | Situations where more control or WAAS is required (e.g. RNAV approaches)    |
| Ease of use                                      | Super Easy                                                                                                                                   | Complexity Rating High                                                      |



# GPSG-1000 GPS/Galileo Constellation Simulator

## GPSG-1000 vs OSPREY

Use the GPSG-1000 for more complex solutions:

- GPS/Galileo Portable 12-Channel Simulator
- WAAS capable
  - Required for RNAV approaches
- GPS signals simulated
  - L1, L1C, L2C, L5
- Galileo signals simulated
  - E1, E5, E5a, E5b, E6
- RAIM / Satellite Health
- Record and Playback feature
- ADS-B application option
  - Required for automated ADS-B and Latency tests when configured with the IFR6000/6015 and IFF45TS
- RCI for automated testing



# GPSG-1000 GPS/Galileo Constellation Simulator

## GPSG-1000 vs OSPREY

